

Q. Draw a circle using Bresenham's circle drawing algorithm, which center (0,0) and radius R=6.

Answer:

Given Data: Center: $(0, 0)$, Radius: $R = 6$

Initialization:

- Starting point, $x_0 = 0, y_0 = 6$.
- Decision Parameter $(P_0) = 3 - 2R$
 $P_0 = 3 - 2 \times 6 = 3 - 12 = -9$

Formulas:

Condition	x_{i+1}	y_{i+1}	P_{i+1}
If $P_i < 0$	$x_i + 1$	y_i	$P_i + 4x_{i+1} + 6$
If $P_i \geq 0$	$x_i + 1$	$y_i - 1$	$P_i + 4(x_{i+1} - y_{i+1}) + 10$

Calculation Table (First Octant):

Step (i)	Previous x (x_i)	Previous y (y_i)	Current P_i	Condition	New x (x_{i+1})	New y (y_{i+1})	Calculation for P_{i+1}	Next P (P_{i+1})
0	0	6	-9	$P < 0$	1	6	$-9 + 4(1) + 6$	1
1	1	6	1	$P \geq 0$	2	5	$1 + 4(2 - 5) + 10$	-3
2	2	5	-3	$P < 0$	3	5	$-3 + 4(3) + 6$	15
3	3	5	15	$P \geq 0$	4	4	$15 + 4(4 - 4) + 10$	25
4	4	4	25	$P \geq 0$	5	3	$25 + 4(5 - 3) + 10$	43
5	5	3	43	$P \geq 0$	6	2	$43 + 4(6 - 2) + 10$	69

Termination Condition:

After Step 5, we get $x_{i+1} = 6$, which is equal to the Radius ($R = 6$). Since x can never be greater than the Radius, the algorithm stops here.

Final Points using 8-Way Symmetry:

(x, y)	$(-x, y)$	$(x, -y)$	$(-x, -y)$	(y, x)	$(-y, x)$	$(y, -x)$	$(-y, -x)$
(0, 6)	(0, 6)	(0, -6)	(0, -6)	(6, 0)	(-6, 0)	(6, 0)	(-6, 0)
(1, 6)	(-1, 6)	(1, -6)	(-1, -6)	(6, 1)	(-6, 1)	(6, -1)	(-6, -1)
(2, 5)	(-2, 5)	(2, -5)	(-2, -5)	(5, 2)	(-5, 2)	(5, -2)	(-5, -2)
(3, 5)	(-3, 5)	(3, -5)	(-3, -5)	(5, 3)	(-5, 3)	(5, -3)	(-5, -3)
(4, 4)	(-4, 4)	(4, -4)	(-4, -4)	(4, 4)	(-4, 4)	(4, -4)	(-4, -4)
(5, 3)	(-5, 3)	(5, -3)	(-5, -3)	(3, 5)	(-3, 5)	(3, -5)	(-3, -5)
(6, 2)	(-6, 2)	(6, -2)	(-6, -2)	(2, 6)	(-2, 6)	(2, -6)	(-2, -6)

Result: Plot these coordinates on graph paper to draw the required circle with center (0,0) and radius 6.

Q. Given the center point coordinates (4, 4) and radius 10, generate all the points to form a circle using the Mid-Point Circle Drawing Algorithm.

Answer:

Given Data:

- Center: $(x_c, y_c) = (4, 4)$
- Radius: $r = 10$

Initialization:

- $x_0 = 0, y_0 = r = 10$
- $P_0 = 1 - r = 1 - 10 = -9$

Formulas:

- If $P_k < 0$: $x_{k+1} = x_k + 1, y_{k+1} = y_k, P_{k+1} = P_k + 2x_{k+1} + 1$
- If $P_k \geq 0$: $x_{k+1} = x_k + 1, y_{k+1} = y_k - 1, P_{k+1} = P_k - 2y_{k+1} + 2x_{k+1} + 1$

Calculation Table:

Step (k)	Prev x_k	Prev y_k	P_k	Condition	New x_{k+1}	New y_{k+1}	Calculation for P_{k+1}	P_{k+1}	$(x_{plot}, y_{plot}) = (x + 4, y + 4)$
0	0	10	-9	$P < 0$	1	10	$-9 + 2(1) + 1$	-6	(4, 14)
1	1	10	-6	$P < 0$	2	10	$-6 + 2(2) + 1$	-1	(5, 14)
2	2	10	-1	$P < 0$	3	10	$-1 + 2(3) + 1$	6	(6, 14)
3	3	10	6	$P \geq 0$	4	9	$6 - 2(9) + 2(4) + 1$	-3	(7, 13)
4	4	9	-3	$P < 0$	5	9	$-3 + 2(5) + 1$	8	(8, 13)
5	5	9	8	$P \geq 0$	6	8	$8 - 2(8) + 2(6) + 1$	5	(9, 12)
6	6	8	5	$P \geq 0$	7	7	$5 - 2(7) + 2(7) + 1$	2	(10, 11)

Termination Condition:

At Step 6, we get $x_{k+1} = 7$ and $y_{k+1} = 7$. Since $x = y$, the first octant is complete. The algorithm stops here.

Corresponding Points using 4-Way Symmetry (Actual Points):

(Formula: $X = x + 4, Y = y + 4$)

(X, Y)	$(-X, Y)$	$(X, -Y)$	$(-X, -Y)$
(4, 14)	(-4, 14)	(4, -14)	(-4, -14)
(5, 14)	(-5, 14)	(5, -14)	(-5, -14)
(6, 14)	(-6, 14)	(6, -14)	(-6, -14)
(7, 13)	(-7, 13)	(7, -13)	(-7, -13)
(8, 13)	(-8, 13)	(8, -13)	(-8, -13)
(9, 12)	(-9, 12)	(9, -12)	(-9, -12)
(10, 11)	(-10, 11)	(10, -11)	(-10, -11)

Conclusion: Plot these coordinates to draw the required circle with center (4,4) and radius 10.

Q. A rectangular clipping window is defined by: $x_{min} = 2$, $x_{max} = 8$ & $y_{min} = 2$, $y_{max} = 6$. A line segment has endpoints: $P_1(1,5)$ & $P_2(7,7)$.

- Compute the region codes for both endpoints using the Cohen-Sutherland method.
- Determine whether the line is: Completely visible or Completely invisible or Partially visible.
- If the line is partially visible, calculate the clipped line segment coordinates.

Answer:

Given Data:

- $x_{min} = 2$, $x_{max} = 8$, $y_{min} = 2$, $y_{max} = 6$
 - Endpoints: $P_1(1, 5)$, $P_2(7, 7)$
-

a) Region Codes (Outcodes):

Cohen-Sutherland Region Code Rules:

- Bit 1 (Top):** 1 if $y > y_{max}$
- Bit 2 (Bottom):** 1 if $y < y_{min}$
- Bit 3 (Right):** 1 if $x > x_{max}$
- Bit 4 (Left):** 1 if $x < x_{min}$

For $P_1(1, 5)$:

- $y = 5$ (Inside vertically) $\rightarrow$ Bit 1: 0, Bit 2: 0
- $x = 1 < x_{min}(2)$ (Left) $\rightarrow$ Bit 3: 0, Bit 4: 1
- Region Code of $P_1 = 0001$**

For $P_2(7, 7)$:

- $y = 7 > y_{max}(6)$ (Top) $\rightarrow$ Bit 1: 1, Bit 2: 0
- $x = 7$ (Inside horizontally) $\rightarrow$ Bit 3: 0, Bit 4: 0
- Region Code of $P_2 = 1000$**

b) Determine line visibility:

- Trivial Acceptance:** Both codes are not 0000. So, not completely visible.
- Trivial Rejection (Bitwise AND):**

$$\begin{array}{r} 0001 \\ 1000 \\ \hline 0000 \end{array}$$

Since the result is 0000, the line is not completely invisible.

Conclusion: The line is **Partially visible**.

c) Clipped line segment coordinates:

Step 1: Slope

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{7 - 5}{7 - 1} = \frac{2}{6} = \frac{1}{3}$$

Step 2: Clip against Left boundary ($x = x_{min} = 2$)

For $P_1(1, 5)$, $x_1 < x_{min}$. Finding new y :

$$y = y_1 + m(x - x_1)$$

$$y = 5 + \frac{1}{3}(2 - 1) = 5 + 0.33 = 5.33$$

New point $P'_1 = (2, 5.33)$

Step 3: Clip against Top boundary ($y = y_{max} = 6$)

For $P_2(7, 7)$, $y_2 > y_{max}$. Finding new x :

$$x = x_1 + \frac{1}{m}(y - y_1)$$

$$x = 1 + 3(6 - 5) = 1 + 3(1) = 4$$

New point $P'_2 = (4, 6)$

Q. A circle has a Center C (0,0) and a Radius $r = 5$. Answer the following questions using the given center and radius.

- Compute the initial decision parameter.
- Calculate the first 6 points in the first octant using the iterative formula.
- Using symmetry, determine all corresponding points of the circle for the computed octant points.

Answer:

Given Data:

- Center: $(x_c, y_c) = (0, 0)$
- Radius: $r = 5$

a) Compute the initial decision parameter:

$$P_0 = 3 - 2r$$

$$P_0 = 3 - 2(5) = 3 - 10$$

$$P_0 = -7$$

b) Calculate the first 6 points in the first octant:

Formulas:

- If $P_k < 0$: $x_{k+1} = x_k + 1$, $y_{k+1} = y_k$, $P_{k+1} = P_k + 4x_{k+1} + 6$
- If $P_k \geq 0$: $x_{k+1} = x_k + 1$, $y_{k+1} = y_k - 1$, $P_{k+1} = P_k + 4(x_{k+1} - y_{k+1}) + 10$

Initialization: $x_0 = 0$, $y_0 = 5$, $P_0 = -7$

Calculation Table (Perfectly Corrected):

Step (k)	Prev x_k	Prev y_k	Current P_k	Condition	New x_{k+1}	New y_{k+1}	Calculation for P_{k+1}	Next P_{k+1}
0	0	5	-7	$P < 0$	1	5	$-7 + 4(1) + 6$	3
1	1	5	3	$P \geq 0$	2	4	$3 + 4(2 - 4) + 10$	5
2	2	4	5	$P \geq 0$	3	3	$5 + 4(3 - 3) + 10$	15
3	3	3	15	$P \geq 0$	4	2	$15 + 4(4 - 2) + 10$	33
4	4	2	33	$P \geq 0$	5	1	$33 + 4(5 - 1) + 10$	59
5	5	1	59	$P \geq 0$	6	0	$59 + 4(6 - 0) + 10$	93

The first 6 points in the first octant are:

$(0, 5), (1, 5), (2, 4), (3, 3), (4, 2), (5, 1)$



c) All corresponding points using symmetry:

যেহেতু Centre $(0, 0)$, তাই আমরা সরাসরি ৮-দিকের প্রতিসাম্য (8-way symmetry) ব্যবহার করব। (এখানে Centre যোগ করার দরকার নেই, কারণ Centre $(0,0)$)

Symmetry Table:

(x, y)	$(-x, y)$	$(x, -y)$	$(-x, -y)$	(y, x)	$(-y, x)$	$(y, -x)$	$(-y, -x)$
(0, 5)	(0, 5)	(0, -5)	(0, -5)	(5, 0)	(-5, 0)	(5, 0)	(-5, 0)
(1, 5)	(-1, 5)	(1, -5)	(-1, -5)	(5, 1)	(-5, 1)	(5, -1)	(-5, -1)
(2, 4)	(-2, 4)	(2, -4)	(-2, -4)	(4, 2)	(-4, 2)	(4, -2)	(-4, -2)
(3, 3)	(-3, 3)	(3, -3)	(-3, -3)	(3, 3)	(-3, 3)	(3, -3)	(-3, -3)
(4, 2)	(-4, 2)	(4, -2)	(-4, -2)	(2, 4)	(-2, 4)	(2, -4)	(-2, -4)
(5, 1)	(-5, 1)	(5, -1)	(-5, -1)	(1, 5)	(-1, 5)	(1, -5)	(-1, -5)

Conclusion: Plot these coordinates to draw the required circle with center $(0,0)$ and radius 5.

Q. A circle is defined with a center C $(3,2)$ & a radius $r = 6$. Now answer to the following question.

- Compute the initial decision parameter for the midpoint circle algorithm.
- Starting from the initial point $(0, r)$, calculate the next 7 points in the first octant.
- Determine the corresponding points in all octants for at least the first 3 computed points.

Answer:

Given Data:

- Center: $(x_c, y_c) = (3, 2)$
- Radius: $r = 6$

a) Initial Decision Parameter:

$$P_0 = 1 - r = 1 - 6 = -5$$

b) Calculate the next 7 points:

Initialization: $x_0 = 0, y_0 = 6, P_0 = -5$

Formulas:

- If $P_k < 0$: $x_{k+1} = x_k + 1, y_{k+1} = y_k, P_{k+1} = P_k + 2x_{k+1} + 1$
- If $P_k \geq 0$: $x_{k+1} = x_k + 1, y_{k+1} = y_k - 1, P_{k+1} = P_k - 2y_{k+1} + 2x_{k+1} + 1$

Calculation Table:

Step (k)	Prev x_k	Prev y_k	P_k	Condition	New x_{k+1}	New y_{k+1}	Calculation for P_{k+1}	P_{k+1}	$(x_{plot}, y_{plot}) = (x + 3, y + 2)$
0	0	6	-5	$P < 0$	1	6	$-5 + 2(1) + 1$	-2	(3, 8)
1	1	6	-2	$P < 0$	2	6	$-2 + 2(2) + 1$	3	(4, 8)
2	2	6	3	$P \geq 0$	3	5	$3 - 2(5) + 2(3) + 1$	0	(5, 8)
3	3	5	0	$P \geq 0$	4	4	$0 - 2(4) + 2(4) + 1$	1	(6, 7)
4	4	4	1	$P \geq 0$	5	3	$1 - 2(3) + 2(5) + 1$	6	(7, 6)

Termination Condition:

At Step 4, we get $x_{k+1} = 5$ and $y_{k+1} = 3$. Although the algorithm stops calculating when $x = y$ (4 = 4), we plot up to Step 4 to get the valid points. The first octant is complete, and the remaining points will be symmetric.

c) Determine corresponding points in all octants for the first 3 computed points:

First 3 computed points (Local): (0, 6), (1, 6), (2, 6)

Symmetry Table (Local Coordinates):

(Centre যোগ করার আগে)

(x, y)	$(-x, y)$	$(x, -y)$	$(-x, -y)$	(y, x)	$(-y, x)$	$(y, -x)$	$(-y, -x)$
(0, 6)	(0, 6)	(0, -6)	(0, -6)	(6, 0)	(-6, 0)	(6, 0)	(-6, 0)
(1, 6)	(-1, 6)	(1, -6)	(-1, -6)	(6, 1)	(-6, 1)	(6, -1)	(-6, -1)
(2, 6)	(-2, 6)	(2, -6)	(-2, -6)	(6, 2)	(-6, 2)	(6, -2)	(-6, -2)

Actual Points (Adding Center (3, 2)):

(Formula: $X = x + 3, Y = y + 2$)

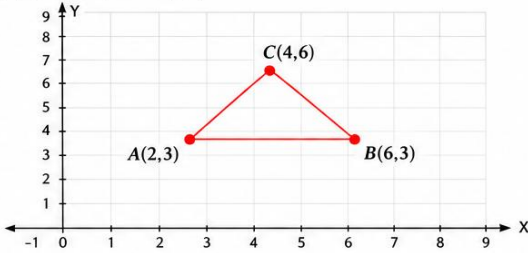
(X, Y)	$(-X, Y)$	$(X, -Y)$	$(-X, -Y)$	(Y, X)	$(-Y, X)$	$(Y, -X)$	$(-Y, -X)$
(3, 8)	(3, 8)	(3, -4)	(3, -4)	(9, 2)	(-3, 2)	(9, 2)	(-3, 2)
(4, 8)	(2, 8)	(4, -4)	(2, -4)	(9, 3)	(-3, 3)	(9, -1)	(-3, -1)
(5, 8)	(1, 8)	(5, -4)	(1, -4)	(9, 4)	(-3, 4)	(9, -2)	(-3, -2)

Conclusion: Plot these coordinates to draw the required circle with center (3,2) and radius 6.

2) TRANSLATION MATH

- 1) Draw a triangle by the three points

$A(2,3)$, $B(6,3)$ and $C(4,6)$.



- 2) Let translation factors $t_x = 2$, $t_y = 2$.

Translation formula:

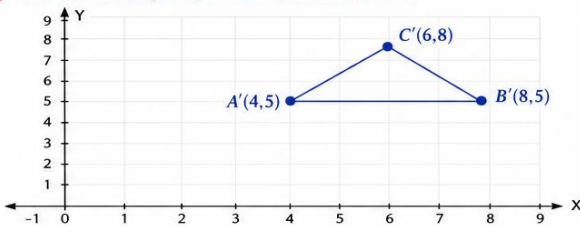
$$x' = x + t_x$$

$$y' = y + t_y$$

Point	Original (x, y)	Calculation	New Point (x', y')
A	(2, 3)	$x' = 2 + 2 = 4$ $y' = 3 + 2 = 5$	$A'(4, 5)$
B	(6, 3)	$x' = 6 + 2 = 8$ $y' = 3 + 2 = 5$	$B'(8, 5)$
C	(4, 6)	$x' = 4 + 2 = 6$ $y' = 6 + 2 = 8$	$C'(6, 8)$

New translation points: $A'(4, 5)$, $B'(8, 5)$, $C'(6, 8)$

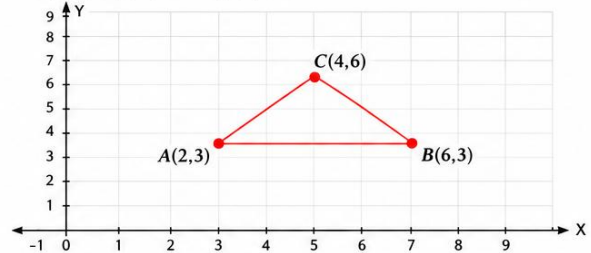
- 3) Draw the new object by the new translation points.



3) SCALING MATH

- 1) Draw a triangle by the three points

$A(2,3)$, $B(6,3)$ and $C(4,6)$.



- 2) Let scaling factors $S_x = 2$, $S_y = \frac{1}{2}$.

Scaling formula:

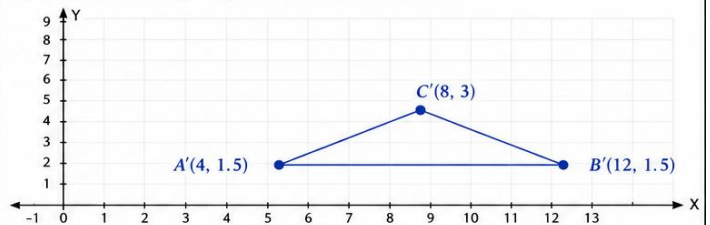
$$x' = x \times S_x$$

$$y' = y \times S_y$$

Point	Original (x, y)	Calculation	New Point (x', y')
A	(2, 3)	$x' = 2 \times 2 = 4$ $y' = 3 \times \frac{1}{2} = 1.5$	$A'(4, 1.5)$
B	(6, 3)	$x' = 6 \times 2 = 12$ $y' = 3 \times \frac{1}{2} = 1.5$	$B'(12, 1.5)$
C	(4, 6)	$x' = 4 \times 2 = 8$ $y' = 6 \times \frac{1}{2} = 3$	$C'(8, 3)$

New coordinate points: $A'(4, 1.5)$, $B'(12, 1.5)$, $C'(8, 3)$

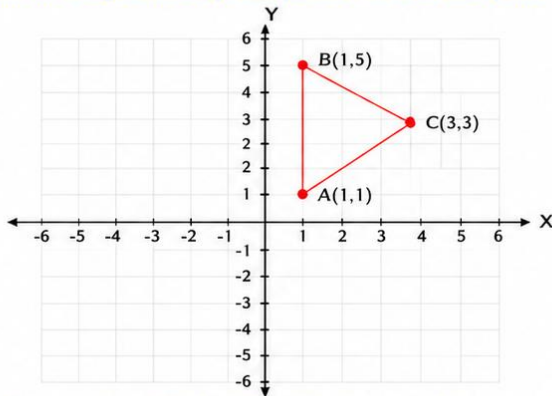
- 3) Draw the new object by the new coordinate points.



Solutions

- (a) Given three points $A(1,1)$, $B(1,5)$, and $C(3,3)$;

1. Draw the triangle formed by these points on the Cartesian plane.



2. Perform a rotation of 90° (anticlockwise) about the origin on the triangle.

Rotation formula (anticlockwise):

$$x' = x \cos \theta - y \sin \theta$$

$$y' = x \sin \theta + y \cos \theta$$

For $\theta = 90^\circ$, $\cos 90^\circ = 0$, $\sin 90^\circ = 1$

So, $x' = -y$, $y' = x$

Now,

For $A(1,1)$:

$$x' = -y = -1$$

$$y' = x = 1 \Rightarrow A'(-1, 1)$$

For $B(1,5)$:

$$x' = -y = -5$$

$$y' = x = 1 \Rightarrow B'(-5, 1)$$

For $C(3,3)$:

$$x' = -y = -3$$

$$y' = x = 3 \Rightarrow C'(-3, 3)$$

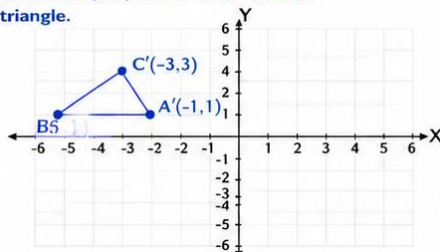
3. Write the new coordinates of A' , B' , and C' after rotation and draw the rotated triangle.

New coordinates:

$$A'(-1, 1)$$

$$B'(-5, 1)$$

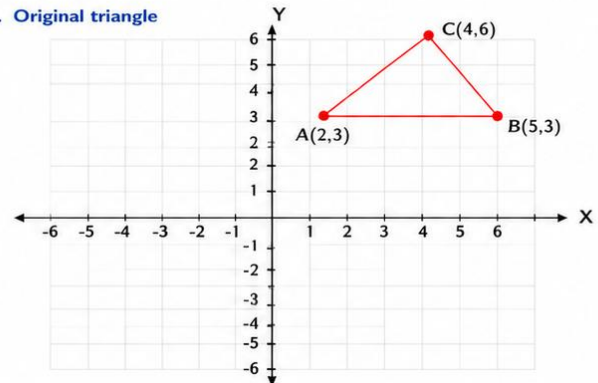
$$C'(-3, 3)$$



- (b) Given a triangle with vertices $A(2,3)$, $B(5,3)$, and $C(4,6)$.

Reflect the triangle about the XY -plane and write the new coordinates of A' , B' , and C' after reflection and draw the reflected triangle.

1. Original triangle



2. Reflection about the XY -plane (i.e., X -axis)

Reflection rule: $(x, y) \rightarrow (x, -y)$

$$A(2,3) \rightarrow A'(2, -3)$$

$$B(5,3) \rightarrow B'(5, -3)$$

$$C(4,6) \rightarrow C'(4, -6)$$

New coordinates:

$$A'(2, -3)$$

$$B'(5, -3)$$

$$C'(4, -6)$$

3. Draw the reflected triangle

